

The team has developed its own antenna that has a radiation pattern, which does not interfere with the sensor's performance. Further, the printed circuit board is provided with its own ground plane and sensor is provided with Faraday's cage or Faraday's shielding to reduce interference. All these steps help minimise interference inside the system.

Smart coding has also helped solve the issues. They studied and understood the way each mobile network behaves with GSM modem (their battery usage, for example) for different types of radiation pattern. This helped design the shielding and coding to operate even in a 900MHz network (as compared to 1800MHz), which is the worst-case scenario.

Keeping the cost low

The main aim of PLUTO was to develop a reliable and cost-effective security solution for consumers. To keep up with this, they had to develop their own mould (metal die) and high-quality stickers with keypads on them.

Since it is a product exclusively designed and manufactured in India, keeping its cost lower than their Chinese counterpart, without compromising on performance, was another major challenge faced. While the security systems available in India with similar features cost about ₹10,000 or more, the developers of B-Safe were able to reduce the cost of the system to ₹6500 by optimising the bill of materials.

Other benefits

The PIR sensor can not only detect security breaches but also the presence of fire and react in a similar fashion. Even though home security was not the initially targeted field of application, B-Safe has also been installed and tested at a home and has worked very well as both intruder alarm and fire detector.

Since the whole system is integrated into a single box, the need for batteries is also reduced to one. The power source used in this system is a digital camera battery, which is easily available these days. Lack of multiple units also makes it very easy for users to install and operate the system without any prior knowledge, and hence serves the targeted rural audience well. You do not require technicians to do this job, thereby avoiding the need for a third person to get involved in the security system of a private area.

Commercially available

It took two years to develop this idea of a low-cost security system to a finished product, said the founders of PLUTO. The team is in the process of patenting the design. Currently, B-Safe is commercially available in the states of Tamil Nadu and Kerala through a couple of vendors, and the company is looking to expand to other places too. ●

The author is a technical correspondent at EFY

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